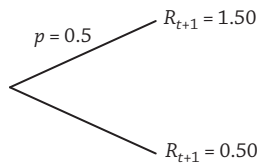


arguing whether you should use pounds or kilograms in your recipe to make apple pie. They represent the same thing; it is a question of which is more useful. What is crucial is the recipe itself—you need the right model for the returns; otherwise it doesn't matter whether you measure your apples in pounds or kilos.

In analyzing portfolios (e.g., as we do in chapter 7), it is easiest to use arithmetic returns because they aggregate when we form portfolios. When we analyze asset returns across time (e.g., as we do in chapters 8 and 11, which focus on long-run equity return predictability and how different assets hedge inflation in the long run, respectively), we use log returns because we can sum them across time. In both instances, we describe the same thing—returns.

This is not to say that whether you use pounds or kilos does not matter. As every dieter knows, you can't switch between calories or kilojoules without the risk of severely wrecking your waistline. Consider the following return process for a gross return:



which is a 50%–50% sequence of winning or losing plus or minus 50% arithmetic returns and the probabilities are independent over time. The investor will eventually lose all her capital investing in this asset. (When you gain 50% and lose 50% you end up losing 25%!) The mean arithmetic return is zero, $E[r_{t+1}] = \frac{1}{2} \times 0.5 + \frac{1}{2} \times -0.5 = 0$ and may give a misleading picture of the long-run return of cumulated wealth. In the long run, your wealth shrinks to zero! The mean log return is

$$E[\bar{r}_{t+1}] = 0.5 \times \ln(1.5) + 0.5 \times \ln(0.5) = -14.38\%,$$

so the negative sign at least indicates that this strategy loses wealth over time. The mean log return is also called the *geometric mean*. (When returns are risky, the arithmetic mean is always greater than the geometric mean.²) But the better way is to look at the full distribution of returns, which is captured by the probability distribution function or density function (see chapter 2). The density function of this lottery is very simple—it provides you with risk (there is dispersion or variance) but zero return. You must have a good reason to invest in the first place,

automatically assume that capital is reinvested. For some references see Cooper (1996), Jacquier, Kane, and Marcus (2003), and Missiakoulis, Dimitrios, and Etriotis (2010).

² An approximate relation is $E[\bar{r}_{t+1}] \approx E[r_{t+1}] - \frac{1}{2}\sigma^2$, where σ^2 is the variance of arithmetic returns. This difference is a Jensen's inequality term, and it drives the rebalancing premium. See chapter 4.